Foundations of Programming

Assignment No.: 22

PROBLEM STATEMENT: Write a menu driven program in C to perform various string operations using in-built functions.

OBJECTIVE:

- To understand the concept of **strings in the C programming language**.
- To learn how to use **built-in string functions** for string manipulation.
- To perform operations such as **length calculation, copying, concatenation, comparison, and reversal** using menu options.

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THEORY:

In the C programming language, a **string** is a sequence of characters terminated by a null character ('\0'). Strings are stored in **character arrays** and manipulated using functions available in the **string.h header file**.

Common **built-in string functions** include strlen() for finding the length of a string, strcpy() for copying one string to another, strcat() for concatenating two strings, and strcmp() for comparing two strings.

A **menu driven program** allows the user to select different operations from a list of options. Based on the user's choice, the corresponding string function is executed. This approach makes the program interactive and easy to use.

PLATFORM: Linux

```
#include <stdio.h>
#include <string.h>
int main()
{
   char str1[100],str2[100],temp[200];
   int choice;
   printf("Enter first string: ");
   scanf("%s",str1);
```

```
    printf("Enter second string: ");
    scanf("%s",str2);
    printf("1.Length\n2.Copy\n3.Concatenate\n4.Compare\n");
    printf("Enter your choice: ");
    scanf("%d",&choice);
    switch(choice)
    {
        case 1:
            printf("Length of first string = %d",strlen(str1));
            break;
        case 2:
            strcpy(temp,str1);
            printf("Copied string = %s",temp);
            break;
        case 3:
            strcpy(temp,str1);
            strcat(temp,str2);
            printf("Concatenated string = %s",temp);
            break;
        case 4:
            if(strcmp(str1,str2)==0)
                printf("Strings are equal");
            else
                printf("Strings are not equal");
            break;
        default:
            printf("Invalid choice");
    }
    return 0;
}
```

```c
#include <stdio.h>
#include <string.h>
int main()
{
    char str1[100],str2[100],temp[200];
    int choice;
    printf("Enter first string: ");
    scanf("%s",str1);
    printf("Enter second string: ");
    scanf("%s",str2);
    printf("1.Length\n2.Copy\n3.Concatenate\n4.Compare\n");
    printf("Enter your choice: ");
    scanf("%d",&choice);
    switch(choice)
    {
        case 1:
            printf("Length of first string = %d",strlen(str1));
            break;
        case 2:
            strcpy(temp,str1);
            printf("Copied string = %s",temp);
            break;
        case 3:
            strcpy(temp,str1);
            strcat(temp,str2);
            printf("Concatenated string = %s",temp);
            break;
        case 4:
            if(strcmp(str1,str2)==0)
                printf("Strings are equal");
            else
                printf("Strings are not equal");
            break;
        default:
            printf("Invalid choice");
    }
    return 0;
}
```

INPUT: Two strings entered by the user.

```
Enter first string: mainfile
Enter second string: backend
1.Length
2.Copy
3.Concatenate
4.Compare
Enter your choice: 3
Concatenated string = mainfilebackend
```

OUTPUT: Displays results of different string operations such as **length, copy, concatenation, and comparison**.

SAMPLE INPUT:
Enter first string: Hello
Enter second string: World
Choice: 3

SAMPLE OUTPUT:
Concatenated string: HelloWorld
CONCLUSION: Thus, the C program successfully demonstrates **different string operations using built-in functions** through a **menu driven approach**, making the program interactive and easy to use.

FAQs:

1. What are **built-in string functions** in C and why are they used?
2. What is the purpose of the **string.h header file** in C programs?
3. How does a **menu driven program improve user interaction** in programming?